

BiGGViz: A Dual-Interface MATLAB Framework for Visualization of Genome-Scale BiGG-based Metabolic Networks

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Abstract

Background: Genome-scale metabolic reconstructions, analyzed with COBRA methods, provide *in silico* models that link genotype to metabolic phenotype. Biochemical, Genetic and Genomic knowledge base (BiGG) models offers curated reconstructions with standardized identifiers, enabling consistent model exchange and integration with downstream tools. Accessible visualization of the resulting reaction and metabolite networks is essential to make these large models interpretable, communicate analyses, and support users in exploring metabolic structure.

Results: BiGGViz is a MATLAB toolbox for metabolic network visualization that provides both scriptable functions for programmatic use and an interactive App for point-and-click analysis. Its main entry point, `exportBiGGViz`, converts COBRA-compatible metabolic models into annotated, flux-aware viewers of reaction and metabolite networks, while the app provides a graphical interface for loading models, switching between reaction and metabolite views, tuning node appearance, applying filters, inspecting local neighborhoods, and exporting the current view. We demonstrate BiGGViz on a panel of BiGG models to show that the toolbox handles diverse network sizes and organisms and serves as a general-purpose visualization layer for COBRA-style workflows.

Conclusion: BiGGViz automates construction of reaction and metabolite network views from COBRA/BiGG models and wraps them in reusable HTML and GUI-based viewers, making exploratory analysis, result communication, and classroom demonstrations of metabolic network structure more standardized and accessible to both coders and non-coders.

Keywords: Genome-scale metabolic models, COBRA, BiGG, network visualization

1 Background

Genome-scale metabolic reconstructions encode the full set of known metabolic reactions, metabolites, and gene-protein-reaction associations for an organism, providing a structural basis for quantitative, constraint-based modeling of cellular metabolism. [1] Within the COntstraint-Based Reconstruction and Analysis (COBRA) framework, these reconstructions are converted into stoichiometric models on which one can impose physicochemical and environmental constraints and then use linear optimization to predict feasible flux distributions and phenotypes. [2, 3] This enables a wide range of system biology studies, including simulation of growth under different nutrient conditions, prediction of gene essentiality, and exploration of metabolic capabilities across strains or species.

The Biochemical, Genetic and Genomic knowledge base (BiGG) provides a curated collection of such genome-scale metabolic models with standardized reactions and metabolite identifiers, consistent annotation, and web-accessible versions of many widely used reconstructions. [4, 5] These standardized BiGG models facilitate model reuse, comparisons across species/strains, and integration with computational toolboxes such as COBRA, which expect a consistent data structure and identifier scheme. [2, 3] In typical COBRA-style workflows [6, 7], users perform flux balance analysis (FBA), flux variability analysis (FVA), or *in silico* gene knockouts on BiGG-derived models to generate predictions about growth, pathway usage, and metabolite production, but interpreting these results ultimately requires understanding how predicted fluxes or perturbations map onto the underlying network and pathway structure.

A range of software already supports metabolic modeling and network visualization, but each addresses a slightly different problem than BiGGViz. The BiGGR package in Bioconductor, for example, focuses on constraint-based modeling and flux visualization in R, providing functions to compute and display flux distributions on small metabolic networks and pathway diagrams rather than general-purpose reaction and metabolite graphs for arbitrary COBRA models. [8] Several MATLAB-based toolboxes, such as SBEToolbox for biological network analysis—their focus is “given a network, analyze its topology” [9], and the Metabolite Network Toolbox for constructing and analysing kinetic or thermodynamic metabolic models, emphasize graph metrics or model dynamics over direct COBRA/BiGG integration [10]; they typically operate on user-specified adjacency structures or SBML models rather than consuming standardized COBRA model structs as their primary input. Other systems, including MetaboNetworks [11] and web components like MetExportViz [12], provide rich exploration of curated pathway or metabolite graphs but are usually tied to specific sources such as KEGG or a dedicated web back end, and are not designed as MATLAB-native front ends that plug directly into existing COBRA workflows.

General network visualization platforms such as Cytoscape offer powerful, extensible environments for graph analysis and visualization, with numerous plugins,

including apps for flux visualization, but they live outside MATLAB and typically require users to export networks and data from COBRA into exchange formats before visualization. [13] Likewise, Escher is a web application and JavaScript library for building and sharing manually curated metabolic pathway maps with data overlays, well suited for subsystem-level, map-based views but not for automatically generating whole-network reaction and metabolite graphs from arbitrary COBRA/BiGG models inside MATLAB. [14] Algorithms such as GIMME in the COBRA Toolbox address a different layer of the workflow by constructing context-specific models from gene expression and a generic reconstruction, producing a pruned COBRA model but leaving visualization to external tools. [15]

From the perspective of a COBRA/BiGG user working in MATLAB, these tools share several practical limitations: they do not provide a one-step conversion from a COBRA struct to both reaction and metabolite network views, they generally lack tight, built-in integration with BiGG identifiers and annotation tables in MATLAB environment, and HTML-based interactive visualizations—when offered—often require additional scripting, external web stacks, or switching to a different software platform. BiGGViz is designed to fill this niche by treating COBRA/BiGG models as its primary input, automatically constructing reaction and metabolite networks, and providing both MATLAB-native and HTML-based interfaces, including interactive controls for coloring, sizing, filtering, pathway highlighting, node- or pathway-seeded neighborhood exploration, and isolating subnetworks corresponding to specific pathways or subsystems, without extra effort in graph construction or web development. In this way, BiGGViz acts as an easy-to-use, real-time visualization layer that supports exploratory analysis and communication of results directly within the COBRA/BiGG ecosystem, while still allowing users to export the resulting networks to external platforms such as Cytoscape when more advanced, platform-specific analyses are needed.

Taken together, these developments highlight a specific gap: there’s no light-weight, COBRA-centric visualization layer that takes a COBRA/BiGG model struct as input and, without additional graph-construction code, produces interpretable reaction and metabolite network views that can be explored interactively inside MATLAB or shared as browser-based visualizations. BiGGViz is designed to address this gap by providing a COBRA/BiGG-specific visualization layer that lowers the barrier to exploratory analysis within MATLAB, while also supporting handoff of network views to external platforms such as Cytoscape when broader downstream analysis is needed. The main objectives are as follows:

- **Provide a minimal wrapper for HTML export.** Given a COBRA-compatible model struct, `exportBiGGViz` builds reaction and metabolite networks and writes interactive HTML viewers.
- **Offer a MATLAB app for interactive exploration.** The BiGGViz app let users switch between reaction and metabolite views, adjust visual encodings, filter dense networks, highlight a selected pathway from a dropdown menu, and inspect local neighborhoods in real time, including pathway-seeded neighborhoods with 1 to 3 hops.

- **COBRA- and BiGG-aware annotations.** BiGGViz works directly with BiGG-style identifiers and COBRA fields so that reaction names, subsystems, formulae and fluxes can be added via simple annotations tables.
- **Optional hand-off to external platforms.** Exported networks, HTML views, and Cytoscape-oriented Excel workbooks can serve as a starting point for further, platform-specific analyses in tools such as Cytoscape.

2 Implementation

2.1 Overall software design

A high-level pipeline diagram (Figure 1) summarizes the workflow from a COBRA/BiGG model and optional annotation tables to the derived reaction and metabolite graphs, interactive HTML viewers, and app-based exports. BiGGViz is organized as a light-weight visualization layer on top of COBRA-style metabolic models in MATLAB. At its core, a set of functions takes a COBRA-compatible `model` struct as input and derive corresponding reaction-reaction and metabolite-metabolite graphs, independent of how the model was reconstructed or loaded (e.g., from BiGG via the COBRA Toolbox). On top of this graph-construction layer, BiGGViz exposes two main interfaces: the scriptable wrapper `exportBiGGViz`, which builds the selected networks and writes self-contained HTML viewers, and the `BiGGViz` app, which reuses the same core but provides a MATLAB GUI for interactive exploration and export.

A key design choice is to keep network structure and annotation metadata separate. The COBRA model defines the stoichiometric network, while optional annotation tables supply names, subsystems, formulae, fluxes, and other attributes that are merged into the visualization without modifying the underlying model. This approach allows BiGGViz to rely on existing COBRA functionality for model I/O and FBA, while focusing its own code on graph construction, visual encodings, and interactive exploration.

2.2 Dependencies and environment

BiGGViz is implemented in MATLAB and builds directly on the COBRA Toolbox for model representation, input/output, and flux analysis. Users are expected to run BiGGViz with a recent MATLAB release (**TODO: R20XX or later; see Availability for tested versions**) and to have the COBRA Toolbox installed and initialized in their MATLAB session. The BiGGViz folder must be added to the MATLAB path so that the core functions and app are available.

BiGGViz operates on COBRA-compatible model structs, such as those provided by the BiGG Models resources in MATLAB (`.mat`) or SBML formats, or by user-specific reconstructions loaded via the COBRA Toolbox. In typical workflows, models are loaded using `readCbModel`, `loadBiggModel`, or equivalent COBRA functions, optionally followed by FBA or related analyses to generate flux distributions that can be passed to BiGGViz as annotation metadata. Development and testing have been performed on Windows 11, but the code itself is platform-independent aside from MATLAB and COBRA Toolbox availability (**TODO: Check for Linux**).

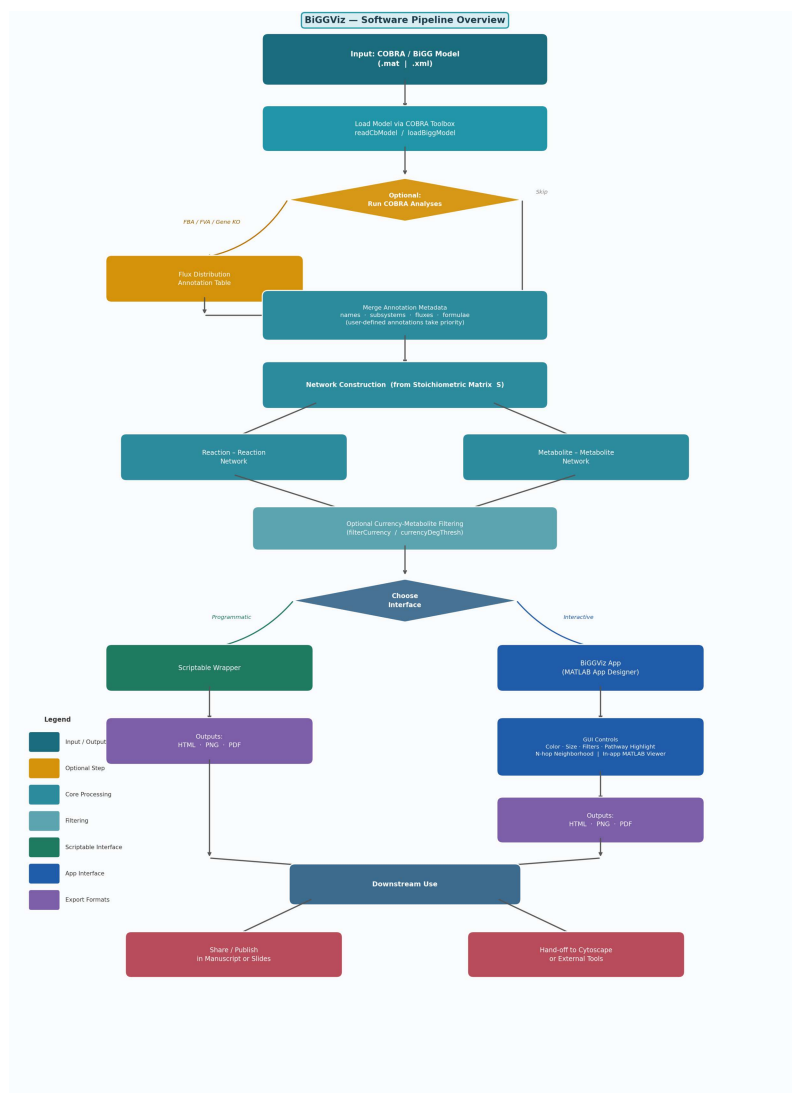


Fig. 1 Overview of the BiGGViz pipeline

2.3 Data model and network construction

BiGGViz interprets the COBRA model as a bipartite reaction-metabolite graph defined by the stoichiometric matrix S , reaction and metabolite identifiers (**rxns**, **met**s), and optional subsystem annotations. All network views are derived from this structure without altering stoichiometry: the bipartite graph is projected into one-mode networks for visualization.

For the reaction network, each reaction in `model.rxns` is a node, and two reactions are connected by an undirected edge when they share at least one metabolite, corresponding to a standard projection onto the reaction layer. For the metabolite network,

each metabolite in `model.mets` is a node, and two metabolites are connected when they participate together in at least one reaction. In both cases, edges are treated as undirected; reversible and irreversible reactions are distinguished through annotation (reversibility flags or flux values) rather than by directed edges, reflecting that the primary goal is structural exploration rather than simulation.

Genome-scale models often contain highly connected “currency” metabolites, such as water, ATP, or cofactors, that create dense hubs and can obscure pathway structure. To mitigate this, BiGGViz includes an optional currency-metabolite filtering mechanism controlled by the `filterCurrency` flag and a degree threshold `currencyDegThresh`, which removes nodes exceeding a user-specified connectivity cut-off from the graph prior to visualization. This simple heuristic reduces clutter and highlights mesoscale organization, at the cost of omitting some generic links; users can disable the filter to recover the full network when needed.

To remain scalable to large constructions, BiGGViz relies on sparse matrix operations when deriving adjacency relationships from the stoichiometric matrix, and constructs only the network(s) requested by the user. Additional scalability controls, such as maximum node degree and minimum component size filters in the app, further limit the number of visible nodes and edges in very dense networks, ensuring that both reaction and metabolite views remain responsive and interpretable even for genome-scale models.

2.4 Wrapper function: `exportBiGGViz`

The `exportBiGGViz` function provides a scriptable entry point for generating network viewers from a COBRA model. Its main inputs are a COBRA-compatible `model` struct, a graph type flag (`'rxn'`, `'met'`, or `'both'`), an optional output directory, an optional reaction-annotation table, and an optional options structure controlling graph-construction settings. The function returns a cell array of file paths to the generated HTML viewers, one per requested network type.

By default, calling `exportBiGGViz(model)` constructs both reaction and metabolite networks, applies standard currency-metabolite filtering, and writes HTML files to the current working directory. Simple variations include requesting only a reaction or metabolite network, disabling currency filtering via the options structure, or supplying an annotation table that includes flux values from an upstream FBA run so that node size or color can reflect reaction fluxes. Internally, `exportBiGGViz` first invokes the core graph-construction routines to derive node and edge lists and then writes these, together with node attributes and layout metadata, into an HTML template that bundles precomputed JSON data and JavaScript code for interactive rendering in a web browser.

2.5 App: BiGGViz interactive MATLAB interface

The BiGGViz app is implemented as a MATLAB App Designer (`.mlapp`) application that wraps the same graph-construction layer in a graphical user interface. The app provides controls for loading a COBRA/BiGG model (from `.mat` or SBML files) and an optional annotation file (`.csv` or `.xlsx`), which is standardized using a

`parseAnnotations` routine so that reaction or metabolite identifiers align with the underlying model. The main panels of the app comprise a control area for data loading and display options, a central visualization panel for the network view, and auxiliary panels for legends and export actions.

Within the app, users can switch between reaction and metabolite network modes, choose coloring schemes (uniform, pathway/subsystem, reversibility, gene association, FBA flux, compartment, or metabolite class), select node size mappings (uniform, degree-based, or flux-based), disable currency filtering via the options structure, and supply an annotation table that may include flux values, node sizes or other user-defined variables that map over existing options. Graph filters such as maximum node degree and minimum component size allow users to suppress hubs or small fragments to simplify dense genome-scale networks, while the neighborhood exploration tools support both node-centered exploration and pathway-seeded exploration with 1 to 3 hops. A pathway highlight dropdown allows users to select a pathway of interest and visually retain that pathway while dimming surrounding nodes, making it easier to inspect how the selected pathway interacts with nearby reactions or pathways. Mode switching between graph-hop and pathway-restricted exploration is handled consistently so that the controls behave predictably across views. A search box (“Find Iout on the corresponding node, and a “Capture View” button enables screen-capture of the current paused frame of the network as seen in the MATLAB App. At any point, the current view can be exported directly from the app as an interactive HTML viewer, as static PNG or PDF figures tuned for sparser, print-friendly layouts, or as a Cytoscape-ready Excel workbook containing separate edge and node tables.

2.6 Implementation quality and testing

BiGGViz is maintained under version control in a public Git repository (TODO) (URL and release tag corresponding to this manuscript are provided in the Availability section), with the MATLAB source, app file, and example scripts stored alongside documentation and a basic test suite (TODO: so far I only have `GettingStarted.mlx`). For validation, BiGGViz was tested on 50 curated BiGG models, with both `.mat` and `.xml` versions evaluated for each model. Across these test cases, the wrapper and app loaded models, constructed the requested network views, and rendered the corresponding interactive outputs as expected. (TODO: maybe also user-supplied small synthetic networks: Additional testing on user-supplied models is used to check robustness of annotation handling, pathway highlighting, neighborhood exploration, and scalability controls). Particular attention was paid to verifying flux-aware visualization: the FBA flux mapping was tested on both BiGG models with native COBRA FBA results and user-supplied FBA flux columns, ensuring consistent behavior across regular COBRA workflows and custom annotations. The project follows general recommendations for bioinformatics software by providing clear installation instructions, a “Getting Started” guide, and reproducible examples, and by using tagged releases to associate specific code versions with figures and results in the manuscript (TODO: Do this after having a Git repository).

TODO: Details of any critical issues and how they were addressed.

3 Results

3.1 Example datasets and loading workflows

To demonstrate BiGGViz on representative COBRA/BiGG workflows, we applied the framework to exemplar genome-scale metabolic models spanning different organisms and reconstruction sizes, including the *Escherichia coli* (E. coli) core model `e_coli_core`, the `iAB_RBC_283` model of human red blood cells, and the `iAF692` model for *Methanosarcina barkeri* strain Fusaro. These examples were selected to illustrate the software across distinct biological systems and annotation structures. Models were loaded from standard COBRA-compatible sources in either MATLAB (`.mat`) or SBML/XML (`.xml`) format, allowing the same visualization workflow to be applied regardless of input file type.

In addition to these representative examples, BiGGViz was tested on 50 curated BiGG models (**TODO: add user-defined models**), using both the MAT and SBML versions of each model. Across these test cases, the wrapper and app successfully loaded the models, constructed the requested reaction and metabolite network views, and generated the expected interactive outputs. These results indicate that BiGGViz is robust to commonly used BiGG/COBRA model formats and can be applied across a range of organisms and network sizes. Figure 2 illustrates representative reaction-network layout for E. coli core model.

3.2 Export of reaction and metabolite network views

For each example model, `exportBiGGViz` generated both reaction-reaction and metabolite-metabolite interactive HTML viewers from the same underlying COBRA model. This provided 2 complementary perspectives on the same reconstruction: reaction-centered views highlighted pathway structure and connectivity among reactions, whereas metabolite-centered views emphasized shared metabolites, compartmentalization, and compound-level neighborhoods.

The exported viewers preserved annotation-aware metadata, including identifiers and optional attributes such as pathway labels, reversibility, gene associations, formulae, compartments, and flux values when available. Because the outputs are written as standalone HTML viewers, they can be shared and explored outside MATLAB without requiring further scripting.

3.3 Flux-aware visualization with COBRA-derived FBA results

To illustrate compatibility with downstream COBRA analyses, flux balance analysis (FBA) results were incorporated into the visualization workflow as optional reaction-level annotations. After computing fluxes with the COBRA Toolbox, the resulting values were passed into BiGGViz and used to enrich reaction-network displays. This enabled visually guided inspection of active vs. inactive regions of the network and made it easier to identify high-flux reactions within dense metabolic graphs.

When the **Color by FBA Flux** option was selected, the viewer displayed a numeric color bar, allowing users to interpret the correspondence between color intensity and flux magnitude directly, in addition to color gradient visual cues. Flux

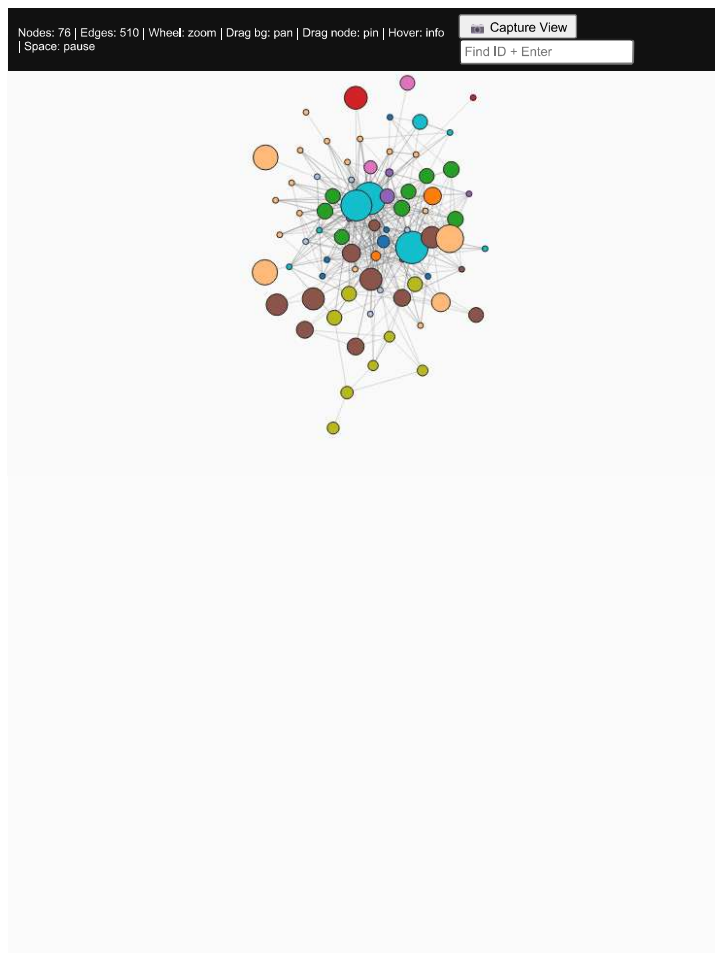


Fig. 2 E. coli Core Reaction Network

information could also be reflected in node-size encodings, providing an additional visual cue for prioritizing reactions during exploratory analysis.

3.4 Interactive pathway and neighborhood exploration in the BiGGViz app

The BiGGViz app supported interactive exploration beyond static export. Users could switch between reaction and metabolite views, adjust node appearance, apply structural filters, and inspect local neighborhoods within the currently displayed network. This made it possible to move from a global view of the reconstruction to local mechanistic exploration without writing additional code.

A dedicated **pathway highlight** feature was included through a dropdown menu that allows the user to select a pathway of interest and retain it visually in the network display. From that selected pathway, the app also supports 1-, 2-, or 3-hop expansion, enabling inspection of how the focal pathway interacts with neighboring reactions, metabolites, or adjacent pathways. This pathway-seeded exploration was useful for tracing cross-pathway relationships while maintaining biological context.

BiGGViz also supports export-oriented workflows intended to facilitate hand-off to Cytoscape-oriented downstream analysis, providing a practical bridge between lightweight MATLAB-native exploration and broader network-analysis environments. This makes BiGGViz useful not only as a standalone viewer, but also as a preprocessing and inspection layer before more specialized platform-specific analysis.